Q5a) Explain the Holdout Method. Differentiate Training Set, Validation Set, and Test Set.

The Holdout Method is a model evaluation technique in which the dataset is divided into separate parts for: Training, Validation, Testing

It helps evaluate how well a machine learning model performs on unseen data.

Working of Holdout Method - The dataset is generally divided as:

Training Set → 70%
Validation Set → 15%
Test Set → 15%

Process of Holdout Method

Step 1: Split Dataset - Dataset is divided into: Training set, Validation set, Test set

Step 2: Train the Model - The training set is used to build the machine learning model.

Step 3: Validate the Model - Validation set is used for: Hyperparameter tuning, Model selection, Performance improvement

Step 4: Test the Model - Final evaluation is done using test set.

Diagram of Holdout Method -

Complete Dataset
↓

Training Set	Validation Set	Testing Set

Advantages of Holdout Method -

- Simple and easy to implement
- Fast evaluation technique
- Suitable for large datasets

Disadvantages - Performance depends on data split, May produce biased results for small datasets

Training Set	**Validation Set**	**Test Set**
Used to train model	Used to tune model	Used to evaluate final model
Largest portion of data	Medium-sized portion	Separate unseen data
Model learns patterns from it	Helps improve performance	Measures final accuracy
Used during learning	Used during optimization	Used after model completion

Example - Suppose dataset contains 1000 records:

Dataset Part	Number of Records
Training Set	700
Validation Set	150
Test Set	150

The Holdout Method is a simple technique used for model evaluation by dividing data into training, validation, and test sets. It helps create reliable and accurate machine learning models.

Q5b) Calculate Accuracy, Precision, Recall, and F1-Score

Given Confusion Matrix ...

Step 1: Identify Values

Term	Value
True Positive (TP)	70
False Negative (FN)	30
False Positive (FP)	20
True Negative (TN)	80

1. Accuracy

Formula - $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$

Calculation = (70 + 80) / (70 + 80 + 20 + 30) = 150 / 200 = 0.75

Accuracy = 75%

2. Precision

Formula - $Precision = \frac{TP}{TP+FP}$

Calculation = 70 / (70 + 20) = 70 / 90 = 0.7778

Precision ≈ 77.78%

3. Recall

Formula - $Recall = \frac{TP}{TP+FN}$

Calculation = 70 / (70 + 30) = 70 / 100 = 0.70

Recall = 70%

4. F1-Score

Formula - $F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

Calculation = (2 × 0.7778 × 0.70) / (0.7778 + 0.70) = 1.0889 / 1.4778 ≈ 0.7368

F1-Score ≈ 73.68%

Accuracy measures overall correctness, Precision measures correctness of positive predictions, Recall measures ability to identify actual positives, and F1-score provides balanced evaluation of Precision and Recall.

Q6a) Explain the Following Text Analysis Steps with Suitable Example

Text Analysis or Text Mining is the process of extracting useful information from textual data using Natural Language Processing (NLP) techniques.

i) Part-of-Speech (POS) Tagging

Part-of-Speech Tagging is the process of assigning grammatical labels to each word in a sentence.

These labels identify whether a word is: Noun, Verb, Adjective, Adverb, Pronoun etc.

Purpose of POS Tagging - Understand sentence structure, Improve text analysis, Support NLP applications

Example – Sentence: "Rahul plays cricket." **POS Tags -**

Word	POS Tag
Rahul	Noun
plays	Verb
cricket	Noun

Applications of POS Tagging – Chatbots, Machine translation, Speech recognition, Sentiment analysis

ii) Lemmatization

Lemmatization is the process of converting words into their base or root form called lemma. It uses vocabulary and grammar rules.

Purpose of Lemmatization - Reduce different word forms, Improve text processing, Normalize textual data

Examples -

Original Word	**Lemma**
Running	Run
Better	Good
Studies	Study
Cars	Car

Example Sentence - "The boys are running fast."

After Lemmatization: "boy be run fast"

POS tagging identifies grammatical roles of words, while lemmatization converts words into their root forms for better text analysis and NLP processing.

Q6b) Use K-Means Clustering and Determine Centroids after One Iteration

Given Data Points - (1,2), (2,1), (3,5), (6,8), (7,6), (5,5)

Initial Centroids: C1 = A(1,1) C2 = B(5,7)

Step 1: Calculate Distance from Each Point to Centroids

Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Distance Table -

Point	**Distance to C1(1,1)**	**Distance to C2(5,7)**	**Cluster**
(1,2)	1	6.40	C1
(2,1)	1	6.71	C1
(3,5)	4.47	2.83	C2
(6,8)	8.60	1.41	C2
(7,6)	7.81	2.23	C2
(5,5)	5.66	2.00	C2

Step 2: Form Clusters

Cluster C1 - (1,2), (2,1)

Cluster C2 - (3,5), (6,8), (7,6), (5,5)

Step 3: Calculate New Centroids

New Centroid for C1: $C1 = \left(\frac{1+2}{2}, \frac{2+1}{2}\right)$

Calculation: C1 = (1.5, 1.5)

New Centroid for C2: $C2 = \left(\frac{3+6+7+5}{4}, \frac{5+8+6+5}{4}\right)$

Calculation: C2 = (5.25, 6)

Final Centroids after One Iteration -

Cluster	New Centroid
C1	(1.5, 1.5)
C2	(5.25, 6)

In K-Means clustering, points are assigned to the nearest centroid, and new centroids are calculated as the mean of cluster points. After one iteration, the updated centroids are: C1 = (1.5, 1.5) & C2 = (5.25, 6)

Q5a) What is Hierarchical Clustering? Explain Hierarchical Clustering Algorithms.

Hierarchical Clustering is an unsupervised machine learning technique used to group similar data objects into clusters in a hierarchical structure.

It creates a tree-like structure called a Dendrogram which shows relationships among clusters.

Dendrogram - A dendrogram is a tree diagram representing hierarchical clustering.

Sample Dendrogram -

```
     ABCD
    /   \
   AB    CD
  / \    / \
 A   B  C   D
```

Characteristics of Hierarchical Clustering - Does not require predefined number of clusters, Produces hierarchy of clusters, Useful for small and medium datasets

Types of Hierarchical Clustering -

1. Agglomerative Clustering (Bottom-Up)
2. Divisive Clustering (Top-Down)

1. Agglomerative Hierarchical Clustering - Agglomerative clustering starts with each data point as a separate cluster and repeatedly merges the closest clusters until one cluster remains.

Steps of Agglomerative Clustering -

Step 1 - Treat each data point as an individual cluster.

Step 2 - Calculate distance between clusters.

Step 3 - Merge the two nearest clusters.

Step 4 - Recalculate distances.

Step 5 - Repeat until all points form one cluster.

Example - A B C D

Initially: {A} {B} {C} {D}

After merging closest clusters: {A,B} {C,D}

Finally: {A,B,C,D}

2. Divisive Hierarchical Clustering

Divisive clustering starts with all data points in one cluster and repeatedly splits clusters into smaller clusters.

Steps of Divisive Clustering -

Step 1 - Place all points into one cluster.

Step 2 - Split cluster into smaller clusters.

Step 3 - Repeat splitting until individual clusters are formed.

Advantages of Hierarchical Clustering - Easy to understand, No need to specify number of clusters, Produces hierarchy of clusters

Disadvantages - Computationally expensive, Difficult for very large datasets, Sensitive to noise

Applications - Document clustering, Image segmentation, Biological classification, Customer segmentation

Hierarchical clustering groups similar data into a hierarchy of clusters using Agglomerative or Divisive approaches. The clustering structure is represented using a dendrogram.

Q5b) k-Fold Cross-Validation

k-Fold Cross-Validation is a model evaluation technique used to measure the performance and generalization ability of machine learning models.

The dataset is divided into: k equal parts called folds

Working of k-Fold Cross-Validation

Step 1 - Split dataset into k equal folds.

Step 2 - Use one fold as testing data and remaining folds as training data.

Step 3 - Train and test the model.

Step 4 - Repeat the process k times by changing testing fold.

Step 5 - Calculate average performance.

Example: 5-Fold Cross-Validation

Fold 1 → Test
Fold 2-5 → Train

Fold 2 → Test
Fold 1,3,4,5 → Train
...

Diagram of k-Fold Cross-Validation -

```
Dataset
 ↓
┌────┬─────┬────┬─────┬────┐
|F1  |F2   | F3 |  F4 |F5  |
└────┴─────┴────┴─────┴────┘
```

One fold → Testing
Remaining folds → Training

Advantages of k-Fold Cross-Validation - Better model evaluation, Reduces overfitting, Efficient use of data, More reliable accuracy

Disadvantages - More computational time, Complex for very large datasets

Applications - Model selection, Performance evaluation, Hyperparameter tuning

k-Fold Cross-Validation is an effective technique for evaluating machine learning models by repeatedly training and testing on different portions of the dataset, producing reliable performance results.

Q6a) What is Text Analysis? Explain the Different Steps Involved in Text Analysis.

Text Analysis or Text Mining is the process of extracting meaningful information and patterns from textual data using Natural Language Processing (NLP), Machine Learning, and Data Mining techniques.

It converts unstructured text into useful insights for analysis and decision-making.

Sources of Text Data – Emails, Social media posts, Reviews, Documents, Chat messages, News articles

Applications of Text Analysis - Sentiment analysis, Spam filtering, Chatbots, Recommendation systems, Document classification

Steps Involved in Text Analysis

1. Text Collection - Gather text data from various sources.

Sources – Websites, Social media, Databases, Documents

2. Text Preprocessing - Clean and prepare text data for analysis.

Activities - Remove punctuation, Remove special characters, Convert to lowercase, Remove stop words

Example - Original Sentence: "The Movie was AWESOME!!"

After preprocessing: movie awesome

3. Tokenization - Splitting text into smaller units called tokens.

Example - Sentence: "I love data science"

Tokens: [I, love, data, science]

4. Stop Word Removal - Remove common words that do not add much meaning.

Examples – is, the, and, of

5. Stemming and Lemmatization

Convert words into root/base form.

Example -

Original Word	**Root Form**
Running	Run
Studies	Study

6. Part-of-Speech (POS) Tagging - Assign grammatical labels to words.

Example -

Word	**POS Tag**
Mayur	Noun
runs	Verb

Diagram of

Text Analysis Process -

Text Collection
↓
Preprocessing
↓
Tokenization
↓
Stop-word Removal
↓
Stemming/Lemmatization
↓
Feature Extraction
↓
Modeling & Analysis
↓
Visualization

7. Feature Extraction - Convert text into numerical representation.

Techniques - Bag of Words, TF-IDF, Word Embeddings

8. Text Analysis / Modeling - Apply machine learning or NLP techniques.

Examples – Classification, Clustering, Sentiment analysis

9. Visualization and Interpretation - Represent insights using: Charts, Word clouds, Dashboards

Text analysis converts unstructured text into meaningful insights using preprocessing, tokenization, feature extraction, and analytical techniques.

Q6b) Social Network Analysis and its Applications

Social Network Analysis is the process of analyzing relationships and interactions among individuals, groups, organizations, or systems using network and graph theory.

In SNA: Nodes represent people or entities. Edges represent relationships or connections.

Example of Social Network -

```
A ---- B ---- C
 \     |
  \    |
    D
```

Where: A, B, C, D → People, Lines → Relationships

Components of Social Network Analysis

Component	Meaning
Node	Person or entity
Edge	Relationship or connection
Network	Collection of nodes and edges

Objectives of Social Network Analysis - Understand relationships, Identify influential users, Analyze communication patterns, Detect communities

Techniques Used in SNA - Graph analysis, Centrality measures, Community detection, Network visualization

Applications of Social Network Analysis -

1. Social Media Analysis - Analyze relationships among users on: Facebook, Twitter, Instagram. Uses - Trend analysis, Influencer identification

2. Recommendation Systems - Suggest friends, products, or content based on network relationships. **Ex -** Friend recommendations on social media.

3. Fraud Detection - Detect suspicious transaction networks in banking systems.

4. Healthcare - Analyze disease spread and patient interaction networks.

5. Marketing and Business - Identify influential customers for targeted advertising.

6. Communication Analysis - Study communication flow within organizations.

7. Cybersecurity - Detect malicious network activities and cyber threats.

Advantages of Social Network Analysis - Better understanding of relationships, Detect hidden patterns, Improves business strategies, Helps identify influencers

Social Network Analysis studies relationships among entities using graph-based techniques. It is widely used in social media, healthcare, fraud detection, marketing, and cybersecurity for discovering useful patterns and connections.

Q5a) K-Means Clustering

Given Dataset ...

Number of clusters: K = 2

Initial Centroids: C1 = (2,3) C2 = (8,6)

Step 1: Calculate Distance of Each Point from Centroids

Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Iteration 1 -

Point	Distance from C1(2,3)	Distance from C2(8,6)	Cluster
A(2,3)	0	6.71	C1
B(4,7)	4.47	4.12	C2
C(3,5)	2.24	5.10	C1
D(6,9)	7.21	3.61	C2
E(8,6)	6.71	0	C2
F(7,8)	7.07	2.24	C2

Formed Clusters - Cluster C1: A(2,3), C(3,5)

Cluster C2: B(4,7), D(6,9), E(8,6), F(7,8)

Step 2: Calculate New Centroids

New Centroid C1: $C1 = \left(\frac{2+3}{2}, \frac{3+5}{2}\right)$

Calculation: C1 = (2.5, 4)

New Centroid C2: $C2 = \left(\frac{4+6+8+7}{4}, \frac{7+9+6+8}{4}\right)$

Calculation: C2 = (6.25, 7.5)

Iteration 2 - Now use new centroids: C1 = (2.5,4) C2 = (6.25,7.5)

Point	Distance from C1	Distance from C2	Cluster
A(2,3)	1.12	6.19	C1
B(4,7)	3.35	2.30	C2
C(3,5)	1.12	4.10	C1

Point	Distance from C1	Distance from C2	Cluster
D(6,9)	6.10	1.52	C2
E(8,6)	5.85	2.30	C2
F(7,8)	6.02	0.90	C2

Clusters Remain Same - Cluster C1: A(2,3), C(3,5)

Cluster C2: B(4,7), D(6,9), E(8,6), F(7,8)

Since clusters do not change, algorithm converges.

Final Centroids

Cluster	Final Centroid
C1	(2.5, 4)
C2	(6.25, 7.5)

Final Cluster Assignment

Cluster C1	Cluster C2
A(2,3)	B(4,7)
C(3,5)	D(6,9)
	E(8,6)
	F(7,8)

Using K-Means clustering with K = 2, the dataset is divided into two clusters. After iterations, the algorithm converges with final centroids: C1 = (2.5, 4) & C2 = (6.25, 7.5)

Q5b) How do you handle Noise and Irrelevant Information in Text Data during Preprocessing?

Text Preprocessing is the process of cleaning and preparing raw textual data before analysis or machine learning.

Text data often contains: Noise, Irrelevant words, Symbols, Duplicate content; Removing such unwanted information improves text analysis accuracy.

Handling Noise in Text Data - Noise refers to unnecessary or meaningless information present in text data. **Examples -** Special symbols, HTML tags, Emojis, Punctuation, Extra spaces

Methods to Handle Noise -

1. Remove Punctuation

Example - Before: "Hello!!! How are you?"

After: Hello How are you

2. Convert to Lowercase

Example - Before: DATA Science

After: data science

3. Remove Special Characters and Numbers

Example - Before: Data@2025!!!

After: Data

4. Remove Extra Spaces

Example - Before: Machine   Learning

After: Machine Learning

Handling Irrelevant Information - Irrelevant information means words that do not add meaningful value to analysis.

1. Stop Word Removal - Remove commonly used words such as: is, the, and, of

Example - Before: "The cat is on the table"

After: cat table

2. Stemming - Reduce words to root form by removing suffixes.

Example – Running -> Run, Playing -> Play

3. Lemmatization - Convert words into meaningful base forms.

Example – Better -> Good, Studies -> Study

4. Tokenization - Split text into smaller units called tokens.

Example - "I love AI"   Tokens: [I, love, AI]

Benefits of Removing Noise and Irrelevant Data - Improves text quality, Increases model accuracy, Reduces processing time, Enhances NLP performance

Bag of Words is a text representation technique that converts text into numerical form based on word frequency.

It ignores: Grammar, Word order and focuses only on occurrence of words.

Example of Bag of Words - Sentence 1: "I love data", Sentence 2: "I love analytics"

Vocabulary - [I, love, data, analytics]

Vector Representation

Sentence	I	love	data	analytics
Sentence 1	1	1	1	0
Sentence 2	1	1	0	1

Advantages of BoW - Simple and easy, Converts text into numerical format, Useful for text classification

Limitations - Ignores word meaning, Ignores context and order

TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF is a statistical technique used to measure the importance of a word in a document relative to a collection of documents.

It gives: High weight to important words, Low weight to common words

TF Formula: $TF = \frac{Number\ of\ times\ term\ appears}{Total\ number\ of\ terms\ in\ document}$

IDF Formula: $IDF = \log\left(\frac{Total\ Documents}{Documents\ containing\ term}\right)$

TF-IDF Formula: $TF\text{-}IDF = TF \times IDF$

Example - If the word "analytics" appears frequently in one document but rarely in others, then its TF-IDF score becomes high.

Applications of TF-IDF - Search engines, Document ranking, Text classification, Recommendation systems

Noise and irrelevant information are removed during preprocessing using techniques like stop-word removal, stemming, and lemmatization. Bag of Words and TF-IDF are important text representation techniques used in text analytics and NLP applications.

Q5a) K-Means Clustering (First Round Only)

Given Points ...

Initial Cluster Centers - Cluster 1 Center = A1(2,10), Cluster 2 Center = B1(5,8), Cluster 3 Center = C1(1,2)

Euclidean Distance Formula - $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 1: Assign Points to Nearest Cluster

Point	Distance to A1(2,10)	Distance to B1(5,8)	Distance to C1(1,2)	Cluster
A1(2,10)	0	3.61	8.06	C1
A2(2,5)	5	4.24	3.16	C3
A3(8,4)	8.49	5	7.28	C2
B1(5,8)	3.61	0	7.21	C2
B2(7,5)	7.07	3.61	6.71	C2
B3(6,4)	7.21	4.12	5.39	C2
C1(1,2)	8.06	7.21	0	C3
C2(4,9)	2.24	1.41	7.62	C2

Step 2: Form Clusters

Cluster 1: A1(2,10)

Cluster 2: A3(8,4), B1(5,8), B2(7,5), B3(6,4), C2(4,9)

Cluster 3: A2(2,5), C1(1,2)

Step 3: Calculate New Cluster Centers

New Center for Cluster 1: (2,10)

New Center for Cluster 2: $\left(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5}\right)$

Calculation:(6,6)

New Center for Cluster 3: $\left(\frac{2+1}{2}, \frac{5+2}{2}\right)$

Calculation: (1.5,3.5)

Final Cluster Centers after First Iteration

Cluster	New Center
Cluster 1	(2,10)
Cluster 2	(6,6)
Cluster 3	(1.5,3.5)

After the first round of K-Means clustering, the points are assigned to the nearest clusters and new cluster centers are recalculated as: Cluster 1 → (2,10), Cluster 2 → (6,6), Cluster 3 → (1.5,3.5)

Q6a) Calculate Accuracy, Precision, Recall and Error Rate for Diabetic Risk

Given Confusion Matrix

Actual Class	**Predicted Yes**	**Predicted No**
Diabetic Risk = Yes	90	210
Diabetic Risk = No	140	9560

Step 1: Identify Values

Parameter	**Value**
True Positive (TP)	90
False Negative (FN)	210
False Positive (FP)	140
True Negative (TN)	9560

1. Accuracy

Formula: $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$ = (90 + 9560) / (90 + 9560 + 140 + 210) = 9650 / 10000 = 0.965

Accuracy = 96.5%

2. Precision

Formula: $Precision = \frac{TP}{TP+FP}$ = 90 / (90 + 140) = 90 / 230 = 0.3913

Precision ≈ 39.13%

3. Recall

Formula: $Recall = \frac{TP}{TP+FN}$ = 90 / (90 + 210) = 90 / 300 = 0.30

Recall = 30%

4. Error Rate

Formula: $Error\ Rate = \frac{FP+FN}{TP+TN+FP+FN}$ = (140 + 210) / 10000 = 350 / 10000 = 0.035

Error Rate = 3.5%

The model has high overall accuracy but low recall and precision for diabetic risk prediction, indicating that many diabetic cases are not correctly identified.

Q5b) Explain the Steps Involved in K-Means Algorithm with Suitable Example

K-Means Algorithm - K-Means is an unsupervised machine learning clustering algorithm used to divide data points into K clusters based on similarity.

Each cluster is represented by a centroid (center point).

Objective of K-Means - Group similar data points together, Minimize distance between points and centroid

Steps Involved in K-Means Algorithm

Step 1: Select Number of Clusters (K)

Choose the number of clusters to be formed.

Example - K = 2 means data will be divided into 2 clusters.

Step 2: Initialize Centroids

Choose initial centroids randomly.

Example - C1 = (1,1) C2 = (5,7)

Step 3: Calculate Distance of Each Point from Centroids

Distance is calculated using Euclidean Distance.

Euclidean Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 4: Assign Points to Nearest Cluster

Each point is assigned to the cluster with minimum distance.

Step 5: Recalculate New Centroids

New centroid is calculated as mean of all points in cluster.

Centroid Formula: $Centroid = \left(\frac{\Sigma x}{n}, \frac{\Sigma y}{n}\right)$

Step 6: Repeat Process

Repeat: Distance calculation, Cluster assignment, Centroid update, until centroids stop changing.

Diagram of K-Means Process -

Select K
↓
Initialize Centroids
↓
Calculate Distance
↓
Assign Clusters
↓
Update Centroids
↓
Repeat Until Convergence

Suitable Example

Given Points - (1,2), (2,1), (5,7), (6,8)

Initial Centroids: C1 = (1,1) C2 = (5,7)

Cluster Assignment

Point	Nearest Cluster
(1,2)	$C1$
(2,1)	$C1$
(5,7)	$C2$
(6,8)	$C2$

New Centroids - Cluster C1: $C1 = \left(\frac{1+2}{2}, \frac{2+1}{2}\right)$ $C1$ = (1.5,1.5)

Cluster C2: $C2 = \left(\frac{5+6}{2}, \frac{7+8}{2}\right)$ $C2$ = (5.5,7.5)

Advantages of K-Means - Simple and fast, Easy to implement, Works well for large datasets

Disadvantages - Need to specify K value, Sensitive to initial centroids, Affected by outliers

K-Means clustering groups similar data points into clusters by repeatedly assigning points and updating centroids until stable clusters are formed.

Q6 a) Define Following Terms with Respect to Confusion Matrix

A Confusion Matrix is a table used to evaluate classification models.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Where: TP → True Positive, TN → True Negative, FP → False Positive, FN → False Negative

i) Accuracy - Accuracy measures overall correctness of the model.

Formula: $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$

Example: If model correctly predicts 90 out of 100 cases: Accuracy = 90%

ii) Precision - Precision measures how many predicted positive cases are actually positive.

Formula: $Precision = \frac{TP}{TP+FP}$

Importance - High precision means fewer false positives.

iii) Recall - Recall measures how many actual positive cases are correctly identified.

Formula: $Recall = \frac{TP}{TP+FN}$

Importance - High recall means fewer false negatives.

iv) AUC-ROC

ROC Curve - ROC (Receiver Operating Characteristic) curve is a graph showing:

- True Positive Rate (Recall)
- False Positive Rate

AUC - AUC (Area Under Curve) measures overall model performance.

Interpretation

AUC Value	Performance
1.0	Perfect model
0.5	Random prediction
< 0.5	Poor model

Importance of AUC-ROC - Evaluates classification models, Compares different models, Measures separability of classes

Accuracy, Precision, Recall, and AUC-ROC are important evaluation metrics used to measure classification model performance.

Q6b) Explain Random Subsampling

Random Subsampling is a model evaluation technique in which dataset is randomly divided multiple times into: Training set & Testing set

The model is trained and tested repeatedly using different random splits.

Working of Random Subsampling -

Step 1 - Randomly split dataset into: Training data, Testing data

Step 2 - Train model using training set.

Step 3 - Evaluate model using testing set.

Step 4 - Repeat process multiple times with different random splits.

Step 5 - Calculate average performance.

Example - Suppose dataset contains 1000 records.

First Split - 700 → Training, 300 → Testing

Diagram of Random Subsampling -

Dataset
↓
Random Split
↓
Training Set + Testing Set
↓
Model Evaluation
↓
Repeat Multiple Times

Second Split - Different random records are selected again.

Advantages - Better performance estimation, Reduces bias, Uses different random samples

Disadvantages - Some data may never be selected, Computationally expensive

Applications - Model evaluation, Performance estimation, Machine learning validation

Random Subsampling repeatedly trains and tests a model on different random splits of data to obtain reliable performance evaluation results.

Q6a) Calculate Accuracy, Precision, Recall and Error Rate for Heart Attack Risk

Given Confusion Matrix -

Actual Class	**Predicted Yes**	**Predicted No**
Heart Attack Risk = Yes	80	220
Heart Attack Risk = No	150	9500

Step 1: Identify Values –

True Positive(TP) – 80, False Negative(FN) – 220, False Positive(FP) – 150, True Negative(TN) - 9500

1. Accuracy: $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$ = (80 + 9500) / (80 + 9500 + 150 + 220) = 9580 / 9950 ≈ 0.9628

Accuracy ≈ 96.28%

2. Precision: $Precision = \frac{TP}{TP+FP}$ = 80 / (80 + 150) = 80 / 230 ≈ 0.3478

Precision ≈ 34.78%

3. Recall: $Recall = \frac{TP}{TP+FN}$ = 80 / (80 + 220) = 80 / 300 ≈ 0.2667

Recall ≈ 26.67%

4. Error Rate: $Error\ Rate = \frac{FP+FN}{TP+TN+FP+FN}$ = (150 + 220) / 9950 = 370 / 9950 ≈ 0.0372

Error Rate ≈ 3.72%

The model has high overall accuracy but lower precision and recall for heart attack risk prediction, meaning many actual heart attack risk cases are not correctly detected.

Q5a) Short Note on Time Series Analysis

Time Series Analysis is a statistical technique used to analyze data collected over time at regular intervals.

It helps identify: Trends, Patterns, Seasonal variations, Future predictions

Examples of Time Series Data - Stock market prices, Weather data, Daily sales records, Temperature readings, Monthly rainfall

Components of Time Series

1. Trend - Long-term increase or decrease in data.

Example - Increase in online shopping sales over years.

2. Seasonal Variation - Regular pattern repeated over fixed intervals.

Example - Higher ice cream sales during summer.

3. Cyclical Variation - Fluctuations occurring over long periods.

Example - Business cycles.

4. Irregular Variation - Random and unpredictable changes.

Example - Natural disasters affecting sales.

Time Series Analysis Process

Collect Data
↓
Visualize Data
↓
Identify Patterns
↓
Build Model
↓
Forecast Future Values

Techniques Used in Time Series Analysis - Moving Average, Exponential Smoothing, ARIMA Model, Trend Analysis

Applications of Time Series Analysis - Weather forecasting, Stock price prediction, Sales forecasting, Traffic prediction, Demand forecasting

Advantages - Helps predict future values, Identifies hidden patterns, Supports business planning

Time Series Analysis is widely used for analyzing time-based data and forecasting future trends in business, finance, healthcare, and weather prediction.